DATA SCIENCE & ARTIFICIAL INTELLIGENCE & CS/IT



Calculus and Optimization

Derivatives and Their Types

Lecture No. 01



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Recap of previous lecture



Topic Taylor & Maclaurin Series

Topics to be Covered



Topic

DERIVATIVES & their Types

(Part 1)

TAYLOR & MACLAURIN SERIES

(MACLAURIN SERIES) → Let $f(x)$ is Continuous func^n defined in a Domain D s.t it's all derivative exist then $f(x)$ can be expanded in the Nbd of 0 $(x \to 0)$ as follows;

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \cdots\cdots\cdots$$

Note ① this series provides Approx value of $f(x)$ in the Nbd of 0

② Coeff of x^n in the M.S.Exp $= \dfrac{f^n(0)}{n!}$

③ In this series we are getting Increasing powers of x

Some Standard Results of Maclaurin Series → All these Results are valid only when $x \to 0$

(1) $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots\cdots$

$\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \cdots\cdots$

(2) $a^x = 1 + x(\ln a) + \frac{x^2}{2!}(\ln a)^2 + \frac{x^3}{3!}(\ln a)^3 + \cdots\cdots$

Gate ✓ $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots\cdots$

(3) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \cdots\cdots$

$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots\cdots = \frac{e^x - e^{-x}}{2}$

(4) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots\cdots$

Taylor Series: If we want to Expand our function $f(x)$ in the Nbd of 'a' then expansion is called T.S.Expansion & it is defined as follows;

$$f(x) = \underbrace{f(a) + (x-a)f'(a)}_{\text{Linear Approximation}} + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots\cdots$$

Note ① this series will provide Approx value of $f(x)$ in the Nbd of a

② Coeff of $(x-a)^n = \dfrac{f^n(a)}{n!}$

③ Here we are getting Increasing powers of $(x-a)$

④ To find Approx Value of $f(x)$ in LINEAR Form, Neglect 2nd & higher degree terms

⑤ " " " " " " Quadratic form, Neglect 3rd " " " "

RECAP: ① T.S Exp of $\sin x^3$ in the Nbd of $x=0$ will be?

$$\sin(x^3) = (x^3) - \frac{(x^3)^3}{3!} + \frac{(x^3)^5}{5!} - \cdots\cdots$$

In the Nbd of $x=0 \Rightarrow x \to 0$ ie x is very small

$\Rightarrow x^3$ will also very very small

or $x^3 \to 0$

② T.S Exp of $\log x$ in the Nbd of $x=1$

$$\log x = \log\left[1+(x-1)\right] = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \cdots\cdots$$

In the Nbd of $x=1 \Rightarrow x \to 1 \Rightarrow (x-1) \to 0$

Q: The function $f(x)=\sqrt{\sin x}$ can be approximated about $\frac{\pi}{6}$ as follows:

(a) ✗ $x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\cdots\cdots$ (in the Nbd of $x=0$ i.e. when $x\to 0$)

(b) ✗ $1/2$ (exact value at $x=\frac{\pi}{6}$)

(c) $\left(x-\frac{\pi}{6}\right)-\frac{\left(x-\frac{\pi}{6}\right)^3}{3!}+\frac{\left(x-\frac{\pi}{6}\right)^5}{5!}-\cdots\cdots$

In the Nbd of $x=\frac{\pi}{6} \Rightarrow x\to\frac{\pi}{6}$

or $\left(x-\frac{\pi}{6}\right)\to 0$

i.e. Angle$(x)\to 0$

i.e. we can't use Standard Result.

(d) ✓ None.

i.e. we will use the Fundamental Concept of T.S.Exp.

$f(x)=\sin x,\ f'(x)=\cos x,\ f''(x)=-\sin x,\ f'''(x)=-\cos x$ ------

$f(\frac{\pi}{6})=\frac{1}{2},\ f'(\pi/6)=\frac{\sqrt{3}}{2},\ f''(\pi/6)=\frac{-1}{2}\quad f'''(\pi/6)=\frac{-\sqrt{3}}{2}$ ----

Now T.S.Exp in the Nbd of $\pi/6$ is :

$$f(x)=f(\pi/6)+(x-\tfrac{\pi}{6})f'(\pi/6)+\frac{(x-\pi/6)^2}{2!}f''(\pi/6)+\text{-------}$$

$$\sin x=\frac{1}{2}+(x-\tfrac{\pi}{6})\left(\frac{\sqrt{3}}{2}\right)+\frac{(x-\pi/6)^2}{2!}\left(\frac{-1}{2}\right)+\text{-------}$$

ie Ans is (d)

Q: The function $f(x)=\sin\left(x-\frac{\pi}{6}\right)$ can be approximated about $\frac{\pi}{6}$ as follows:

(a) $x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\cdots\cdots$

(b) $1/2$

(c) $\left(x-\frac{\pi}{6}\right)-\frac{\left(x-\frac{\pi}{6}\right)^3}{3!}+\frac{\left(x-\frac{\pi}{6}\right)^5}{5!}-\cdots\cdots$

(d) None.

$\Downarrow$

$x \to \frac{\pi}{6}$

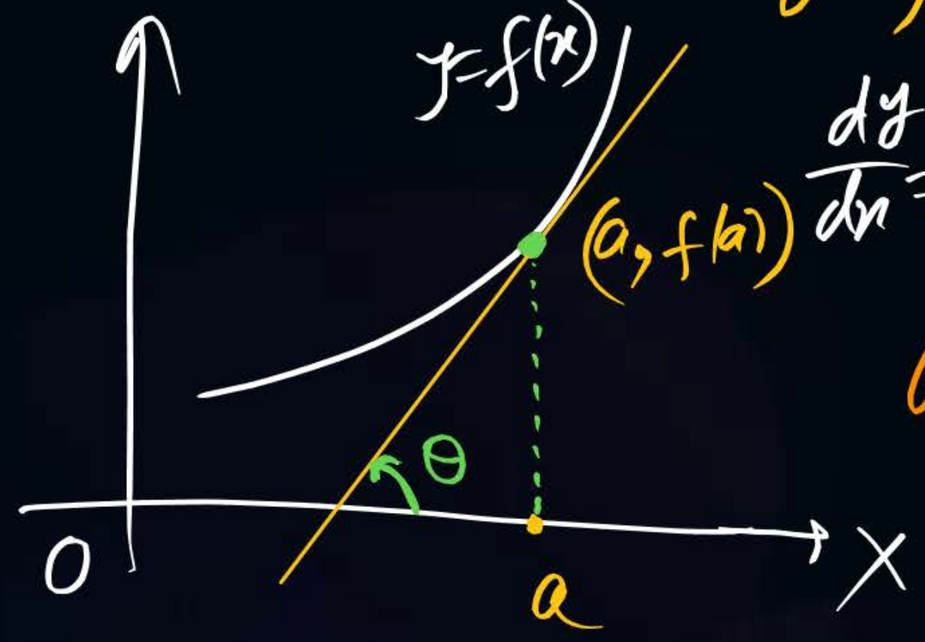
or $\left(x-\frac{\pi}{6}\right)\to 0$

or Angle $\to 0$

i.e. we can apply Standard Result of Maclaurin series.

DIFFERENTIABILITY

Geometrical Meaning →



for $y=f(x)$:

$\frac{dy}{dx} = \boxed{f'(x) = \tan\theta}$ = slope of tangent at any Random point x

for eg at $x=a$, slope of tangent $= f'(a)$

& at $x=b$, " " " $= f'(b)$

& so on - - - - -

Slope of this tangent $= \tan\theta$

& equⁿ of tangent at $(a, f(a))$ is :

$$\boxed{y - f(a) = f'(a)(x-a)}$$

Mathematical Defⁿ — $f(x)$ is said to be differentiable at $x=a$, if it is possible to find the value of $\lim_{x\to a}\left(\frac{f(x)-f(a)}{x-a}\right)$ & value of this limit is called Derivative of $f(x)$ at $x=a$ & it is denoted by $f'(a)$ i.e

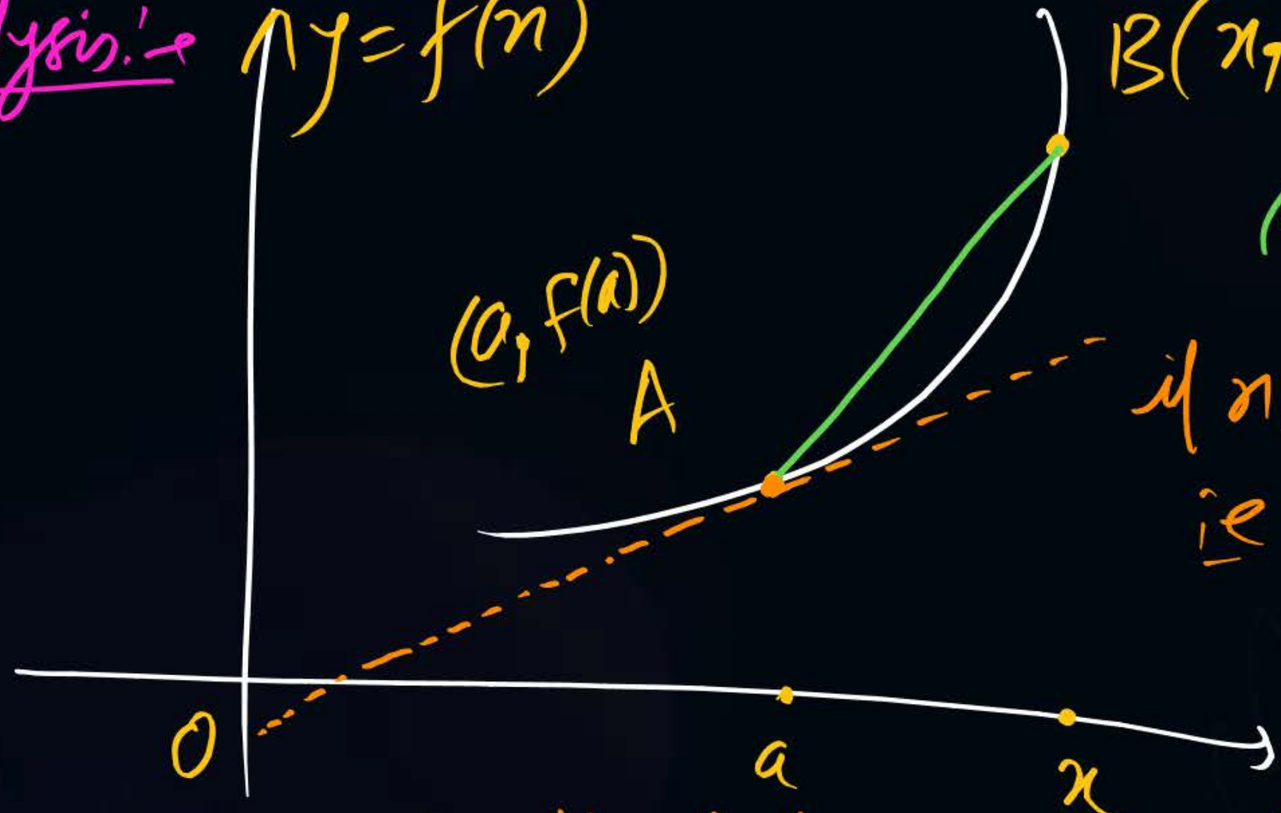
$$\boxed{f'(a)=\lim_{x\to a}\left(\frac{f(x)-f(a)}{x-a}\right)}$$

$$\text{LHD}=\lim_{h\to 0}\left(\frac{f(a-h)-f(a)}{-h}\right)$$

$$\text{RHD}=\lim_{h\to 0}\left(\frac{f(a+h)-f(a)}{h}\right)$$

When LHD = RHD then $f(x)$ is said to be Differentiable at $x=a$

Analysis:-



Slope of line $AB = \dfrac{f(x)-f(a)}{x-a}$

if x lies in the Nbd of a

i.e x lies very-very close to a

i.e $x \to a$ then

Chord will convert into tangent

$\therefore$ Slope of tangent = Slope of Chord (AB)

$$f'(a) = \lim_{x \to a} \left(\frac{f(x)-f(a)}{x-a} \right)$$

Note ① Continuity is the Necessary Condition for Differentiability.

But it is not Sufficient Condition.

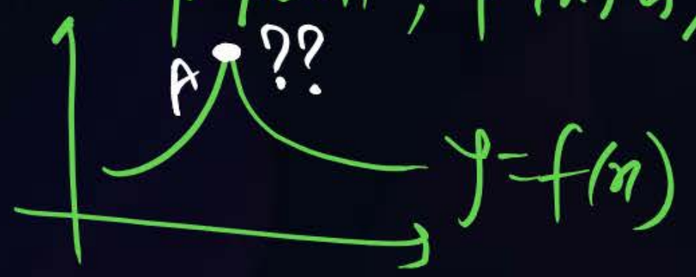
② N Cond^n is ⟶ Continuity

Sufficient Cond^n is ⟶ $\lim_{x \to a} \left(\frac{f(x)-f(a)}{x-a} \right)$ must exist.

Q: Differentiability ⟹ Continuity or Continuity ⟹ Differentiability

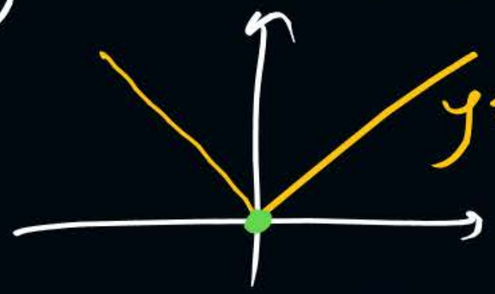
True false.

③ At sharp point, f(x) is said to be Non Differentiable.

eg 

at A, f(x) is Non Diff, ∵ it is not possible to find unique tangent at A.

eg At $x=0$, $f(x)=|x|$ is Continuous but Not Differentiable.



(4) Shortcut Method of finding LHD and RHD of $f(x)$ →

W.K.T that $LHL = \lim_{x \to a^-} f(x)$ & $RHL = \lim_{x \to a^+} f(x)$

Similarly, LHD of $f(x)$ = LHL of $f'(x)$ & RHD of $f(x)$ = RHL of $f'(x)$

$= \lim_{x \to a^-} f'(x)$ $= \lim_{x \to a^+} f'(x)$

Q: Find the Derivative of $y = |x|$?

Sol: $\frac{dy}{dx} = \frac{d}{dx}|x| = \frac{d}{dx}(\sqrt{x^2}) = \frac{d}{dx}(x^2)^{1/2} = \frac{1}{2}(x^2)^{\frac{1}{2}-1} \cdot \frac{d}{dx}(x^2) = \frac{1}{2}(x^2)^{-1/2}(2x)$

$= \frac{x}{\sqrt{x^2}} = \frac{x}{|x|} = \frac{x|x|}{|x|^2} = \frac{x|x|}{x^2} = \frac{|x|}{x}$, $x \neq 0$

i.e $\boxed{\frac{d}{dx}|x| = \frac{|x|}{x}, \; x \neq 0}$ Learn by ❤️

Q: Find LHL & RHD of $f(x)=|x|$ at $x=0$

Sol:
$$f(x)=|x|=\begin{cases}-x, & x<0\\ +x, & x\geq 0\end{cases}$$

$$f'(x)=\begin{cases}-1, & x<0\\ 1, & x>0\end{cases}$$

LHD of $f(x)=f'(0^-)=-1$

RHD of $f(x)=f'(0^+)=+1$

$\because$ LHD $\neq$ RHD so $f(x)=|x|$ is not Differentiable at $x=0$

Note- ① $\frac{d}{dx}(|x|)=\frac{|x|}{x}=\begin{cases}-1, & x<0\\ \text{DNE}, & x=0\\ +1, & x>0\end{cases}$ i.e $f'(0)=\text{DNE}$, LHD $=-1$, RHD $=+1$

② Also find $f'(5)=?=1$, $f'(7)=?=1$, $f'(-4)=-1$

Q: At $x=0$, $f(x)=|x|^3$ is

(a) once Diff but not twice

(b) twice " " " thrice ✓

(c) thrice Diff

(d) Not Differentiable

$$f(x)=|x|^3=\begin{cases}-x^3, & x<0\\ +x^3, & x>0\end{cases}$$

$$f'(x)=\begin{cases}-3x^2, & x<0\\ 3x^2, & x>0\end{cases}$$ LHD = 0, RHD = 0 ✓

$$f''(x)=\begin{cases}-6x, & x<0\\ 6x, & x>0\end{cases}$$ LHD = 0, RHD = 0 ✓✓

$$f'''(x)=\begin{cases}-6, & x<0\\ 6, & x>0\end{cases}$$ LHD = −6, RHD = 6

$\because$ LHD $\neq$ RHD for 3rd Derivative of $f(x)$ = DNE

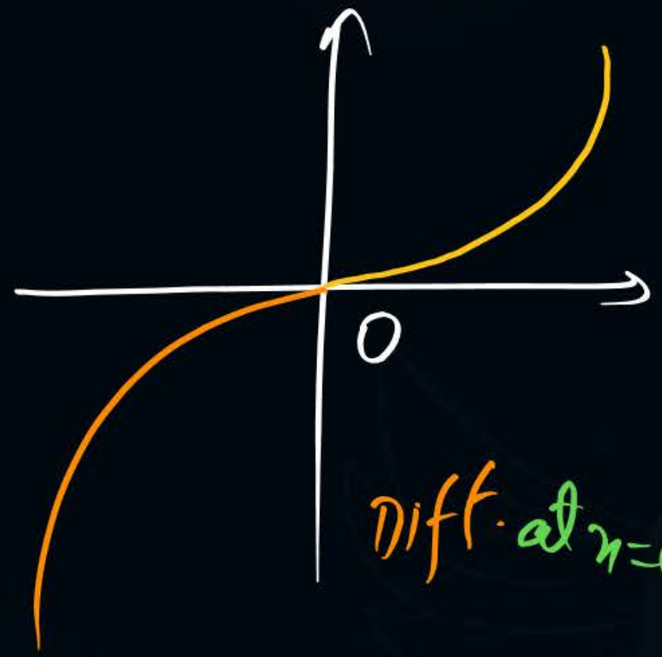
(M-II) Using Graph :-

$y = f(x) = |x|^3 = \begin{cases} -x^3, & x < 0 \\ x^3, & x \geq 0 \end{cases}$



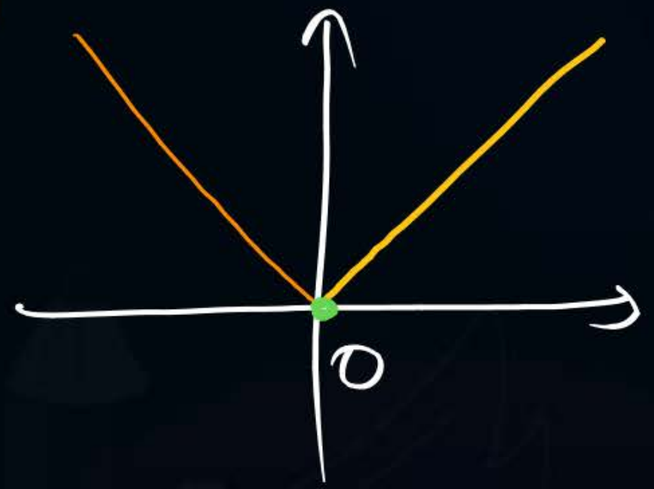
is Diff. at $x = 0$

$f'(x) = \begin{cases} -3x^2, & x < 0 \\ 3x^2, & x \geq 0 \end{cases}$



Diff. at $x = 0$

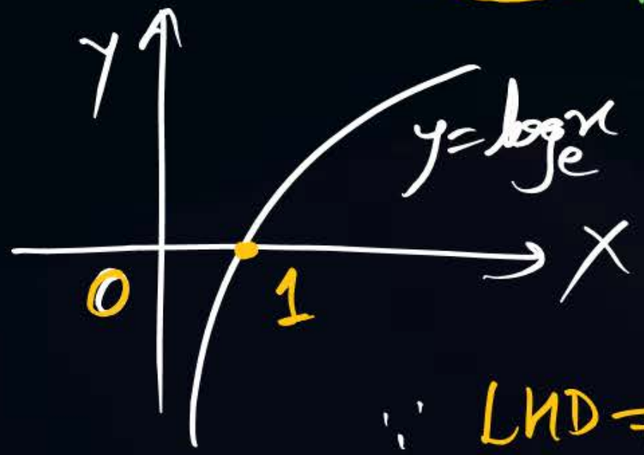
$f''(x) = \begin{cases} -6x, & x < 0 \\ 6x, & x \geq 0 \end{cases}$



Not further Diff at $x = 0$

Q. At $x=1$, $f(x)=|\log_e x|$ is _____

(M-I) $y=f(x)=|\log_e x|=\begin{cases}-\log_e x, & 0<x\le 1\\ +\log_e x, & x>1\end{cases}$ $\Rightarrow f'(x)=\begin{cases}-\frac{1}{x}, & 0<x<1\\ \frac{1}{x}, & x>1\end{cases}$



At $x=1$ → LHD of $f(x)=f'(1^-)=-1$

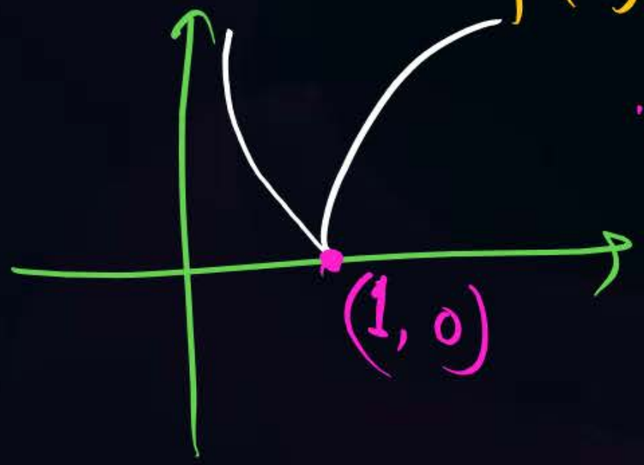
RHD of $f(x)=f'(1^+)=+1$

$\because$ LHD $\neq$ RHD so $f(x)=|\log x|$ is Not Diff at $x=1$

$\because$ F Sharp point at $x=1$

& it is Non Diff there.

(M-II) $y=|\log x|$



Explanation:

$\log_e(4) = ? = 2\log_e 2 = +ve$

$\log_e\left(\frac{1}{4}\right) = ? = \log_e(2)^{-2} = -2\log_e 2 = -ve$

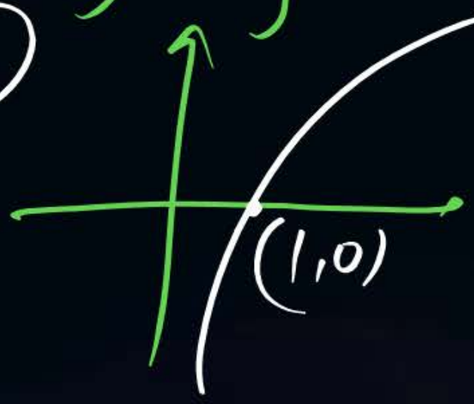
i.e. $y = \log_e x$ is -ve when $0 < x < 1$

& $y = \log x$ is +ve $\therefore$ $x > 1$

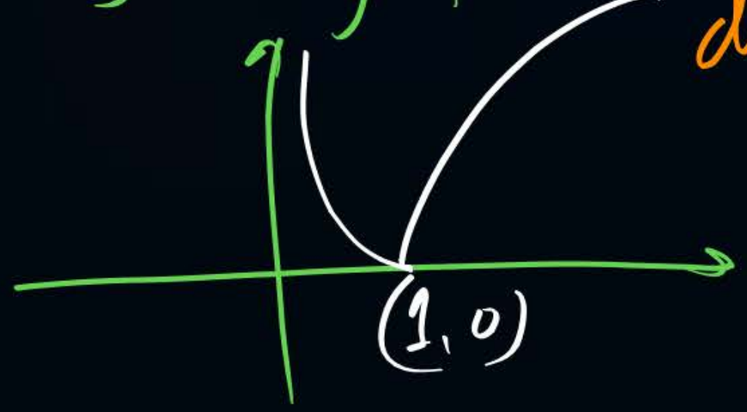


Note:

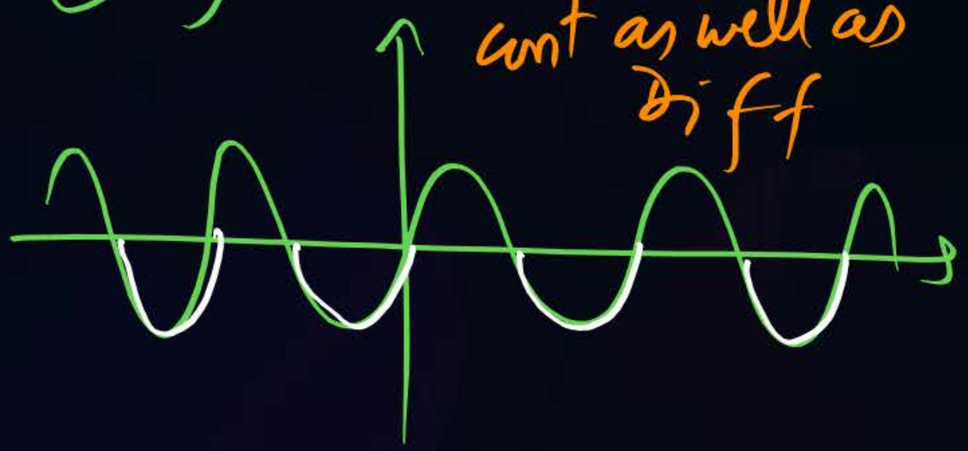
(1) $y = \log x$ Cont at Diff both



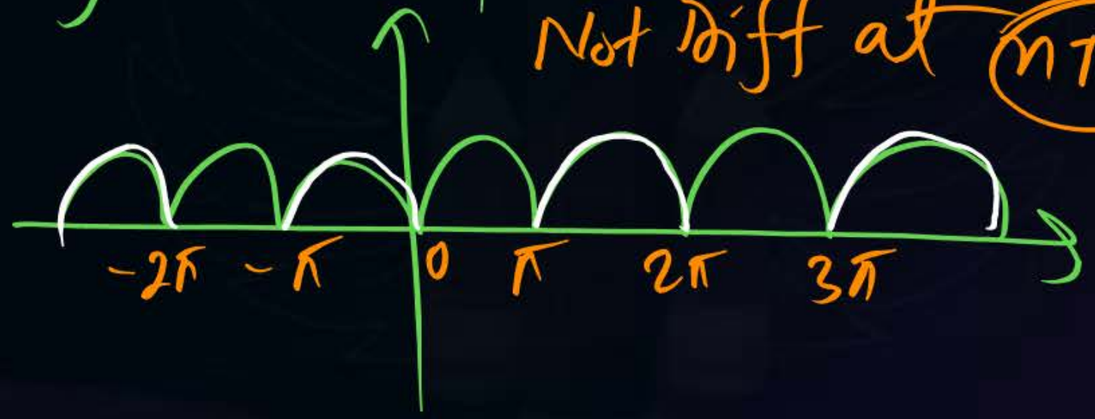
$y = |\log x|$ Cont but not diff at $x=1$



(2) $y = \sin x$ Cont as well as Diff



$y = |\sin x|$ Continuous But Not Diff at $n\pi$



Q: if $f(x)=\begin{cases} e^x & , x<1 \\ ax^2+bx+\log x & , x\geq 1 \end{cases}$ is Differentiable everywhere then find a & b ?

Sol: $\because f(x)$ is Continuous at $x=1$, so

$$\text{LHL} = \text{RHL} = f(1)$$

$$\lim_{x\to 1^-}(e^x) = \lim_{x\to 1^+}(ax^2+bx+\log x) = a+b$$

$$e = a+b = a+b \quad \text{---(1)}$$

By solving (1) & (2), $a=-1$, $b=e+1$

$f'(x)=\begin{cases} e^x & , x<1 \\ 2ax+b+\frac{1}{x} & , x>1 \end{cases}$

$\because f(x)$ is Diff at $x=1$ so

$$\text{LHD of } f(x) = \text{RHD of } f(x)$$

$$\text{LHL of } f'(x) = \text{RHL of } f'(x)$$

$$(e^x)_{x=1^-} = \left(2ax+b+\frac{1}{x}\right)_{x=1^+}$$

$$e = 2a+b+1 \quad \text{---(2)}$$

Note:

Curve: $y = f(x)$



Surface: $z = f(x, y)$



Types of Derivatives →

Ordinary Derivative → exist in case of curve i.e. $y = f(x)$

Partial Derivative → " " " surface i.e. $z = f(x, y)$

Total Derivative → if $z \to (x, y) \to t$ (alone)

Derivative by using chain Rule → $z \to (x, y) \to (r, s)$

Derivative by using Jacobian Concept → if $(u, v) \to (x, y)$

then $\frac{\partial(u,v)}{\partial(x,y)}$ is called Jacobian

THANK - YOU

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